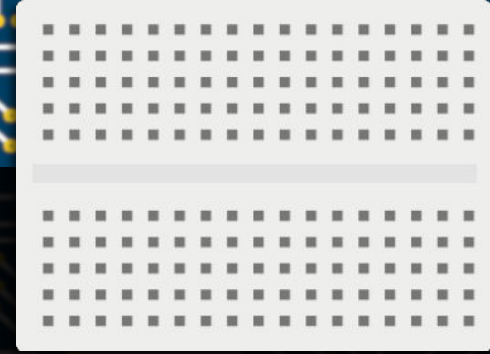
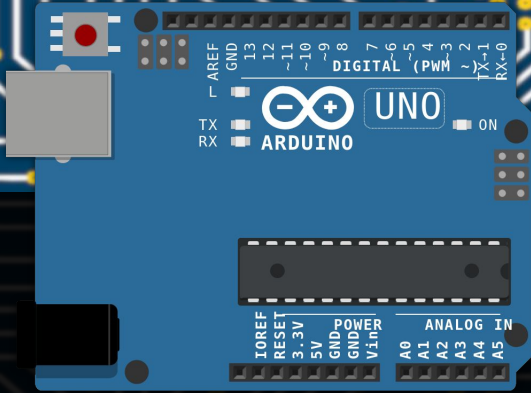
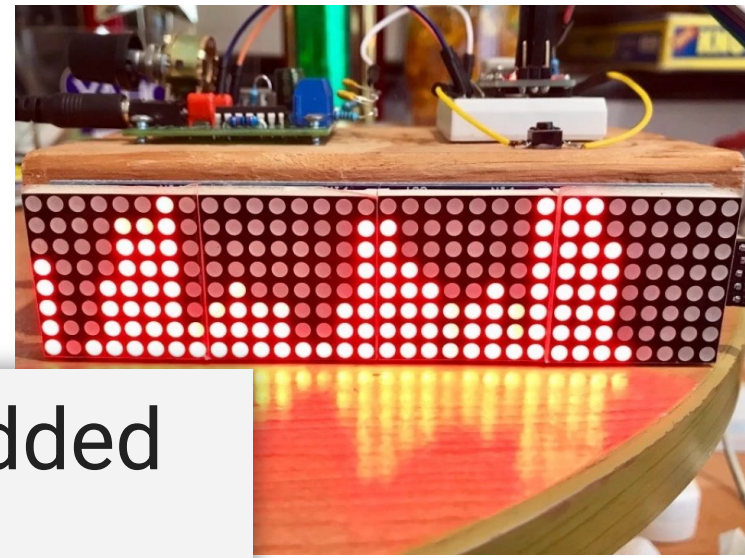
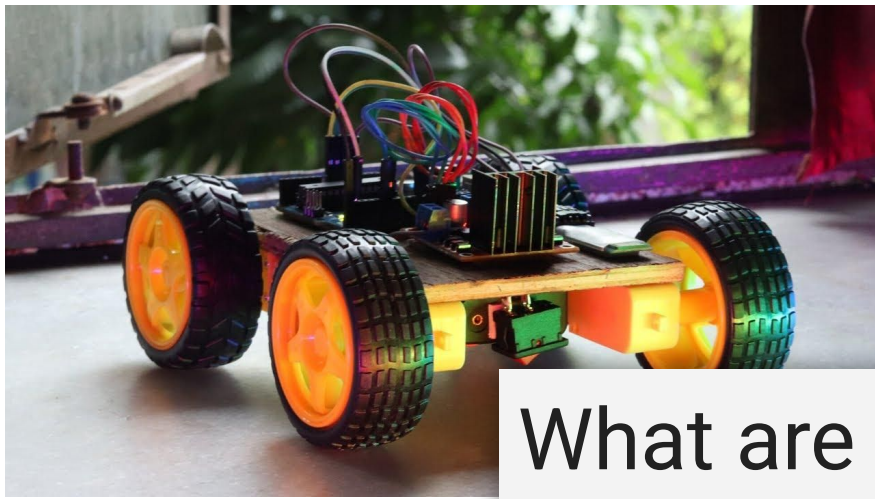


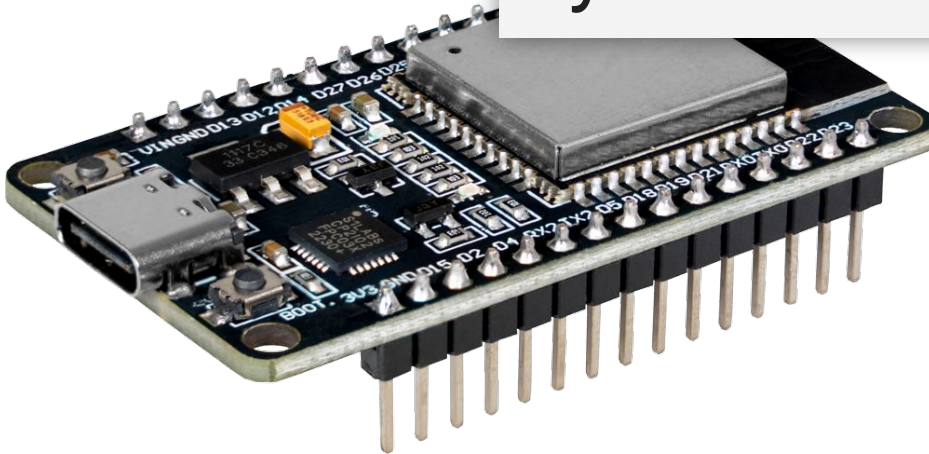


Arduinos!

Flipped Session

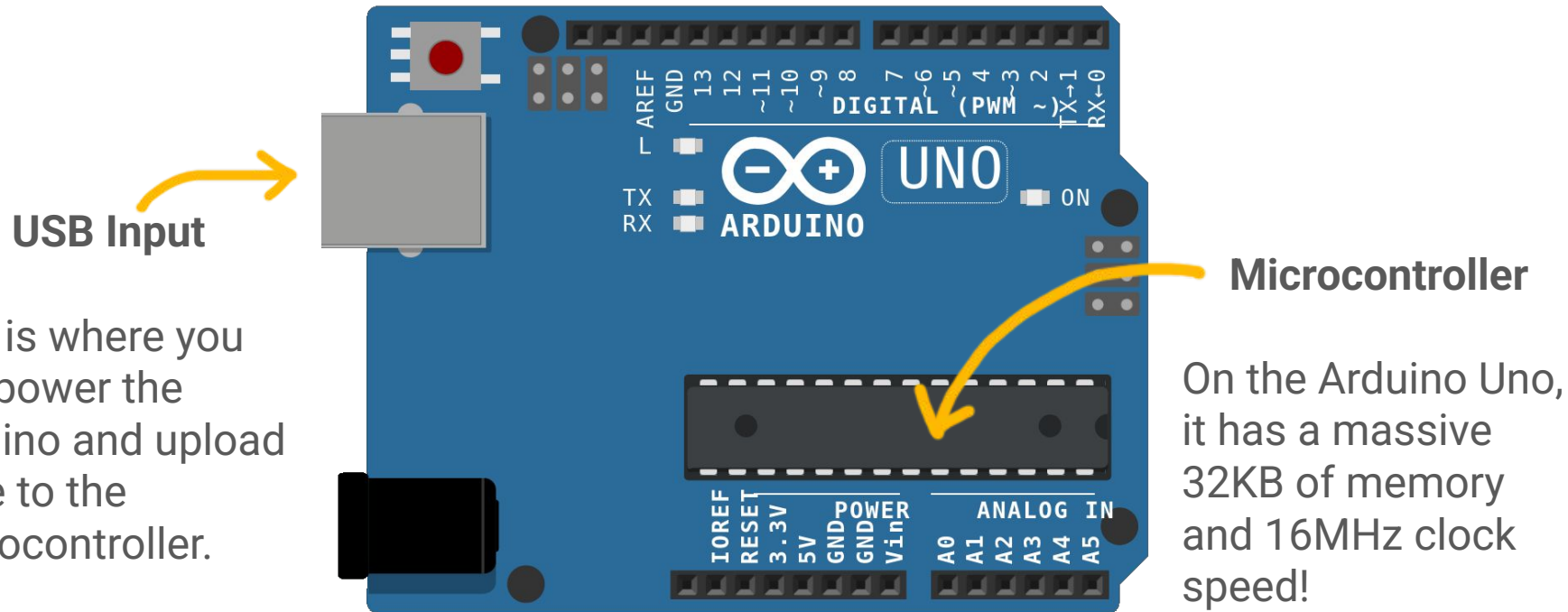


What are Embedded Systems?



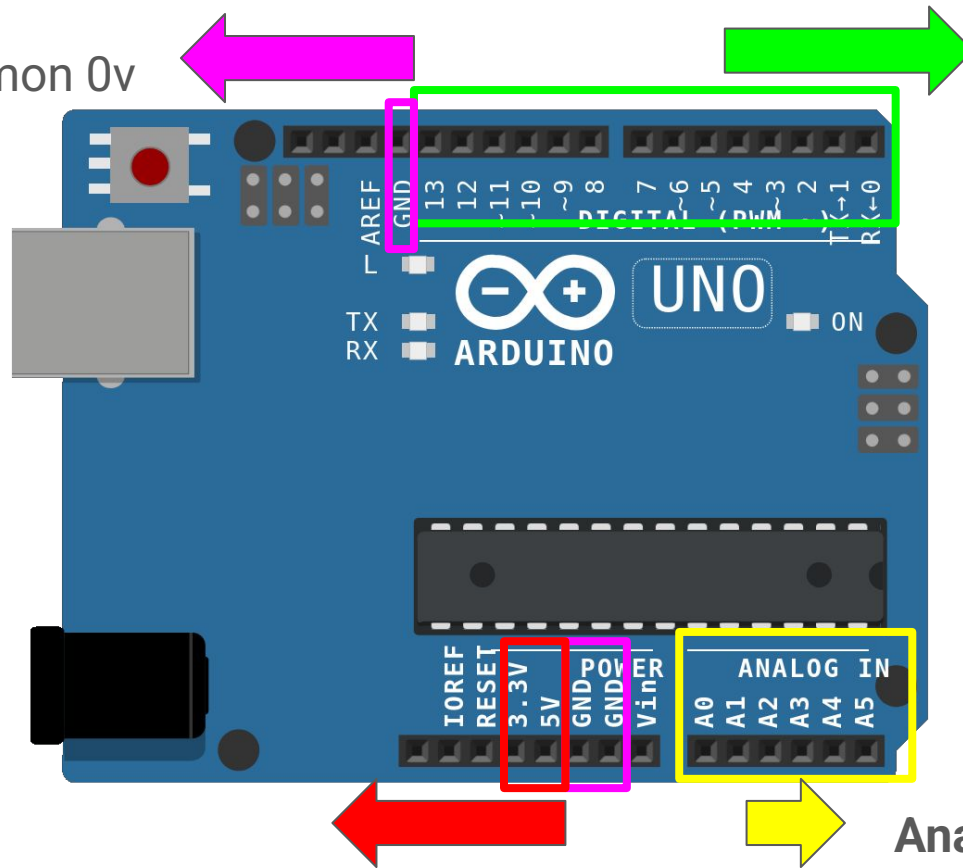
Many Products Depend on Embedded Systems

Arduinos are mini computers which send and receive electronic signals over a set of pins. There are many types, but we're using the Arduino Uno.



Ground

This is the common 0v reference point.



Digital IO

These can take digital inputs, or output a digital signal by being turned on or off by the program.

Pins have different features (EG: PWM)

Power

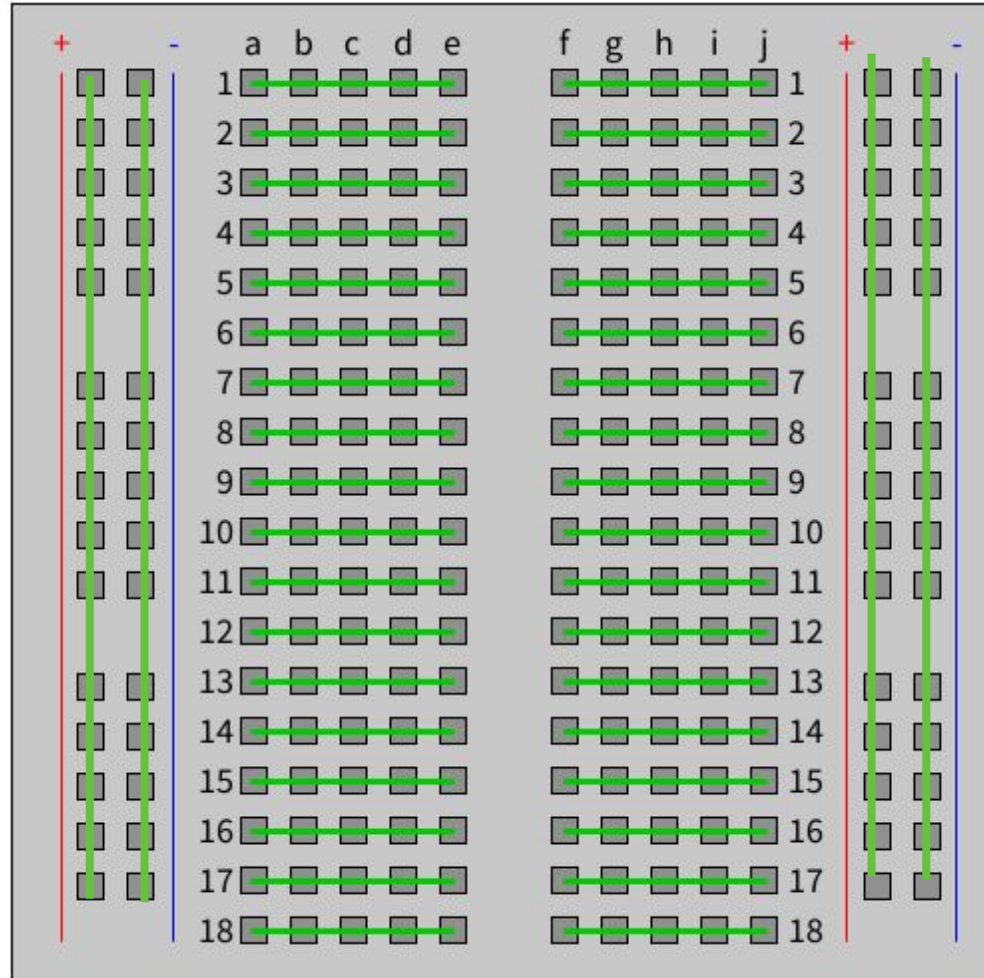
These are your power pins. We have 3.3v and 5v - **Only use 3.3V today!**

Analogue input

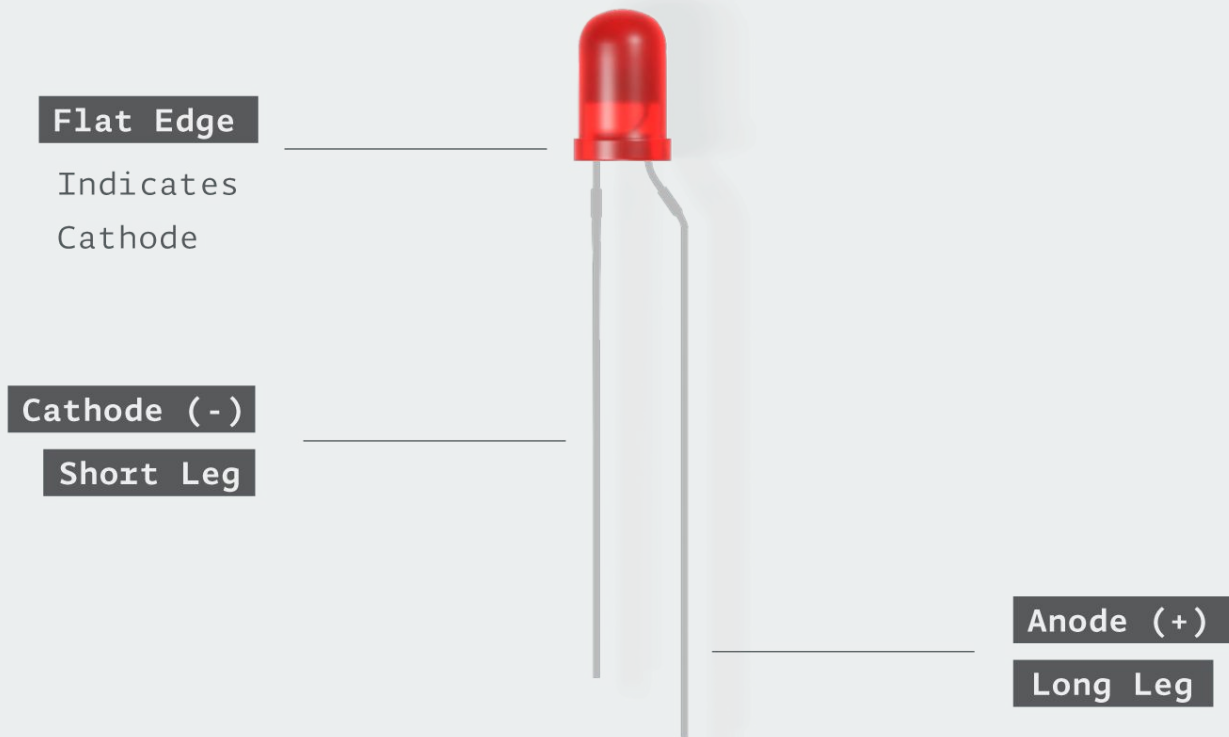
These can take input and measure the voltage (0-5v)

Breadboards!

They're
connected
along green
lines!



Polarity - it matters!

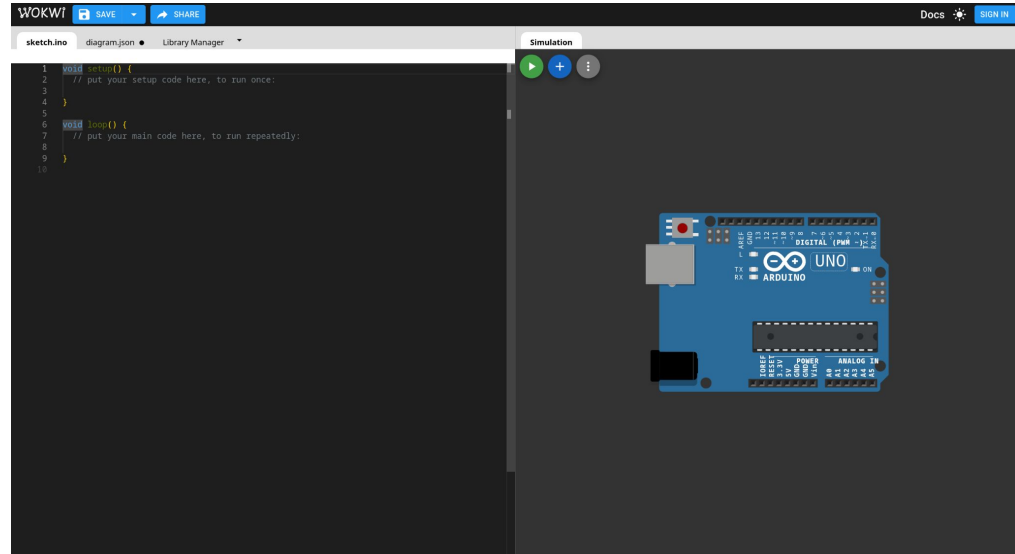


Wowki/TinkerCAD

You can use arduinos at our chill coding session!!!!!!!

Wowki and TinkerCAD allow you to emulate an arduino and create electronics projects at home.

If we don't have enough arduinos try using one of these tools to test your code out!



<https://wokwi.com/projects/new/arduino-uno>

Let's get started!

Get started on challenges on the slides, going at your own pace! Click the links on the right to navigate between tasks!



New to arduino?
Start Here!

Simple logic gates with LED

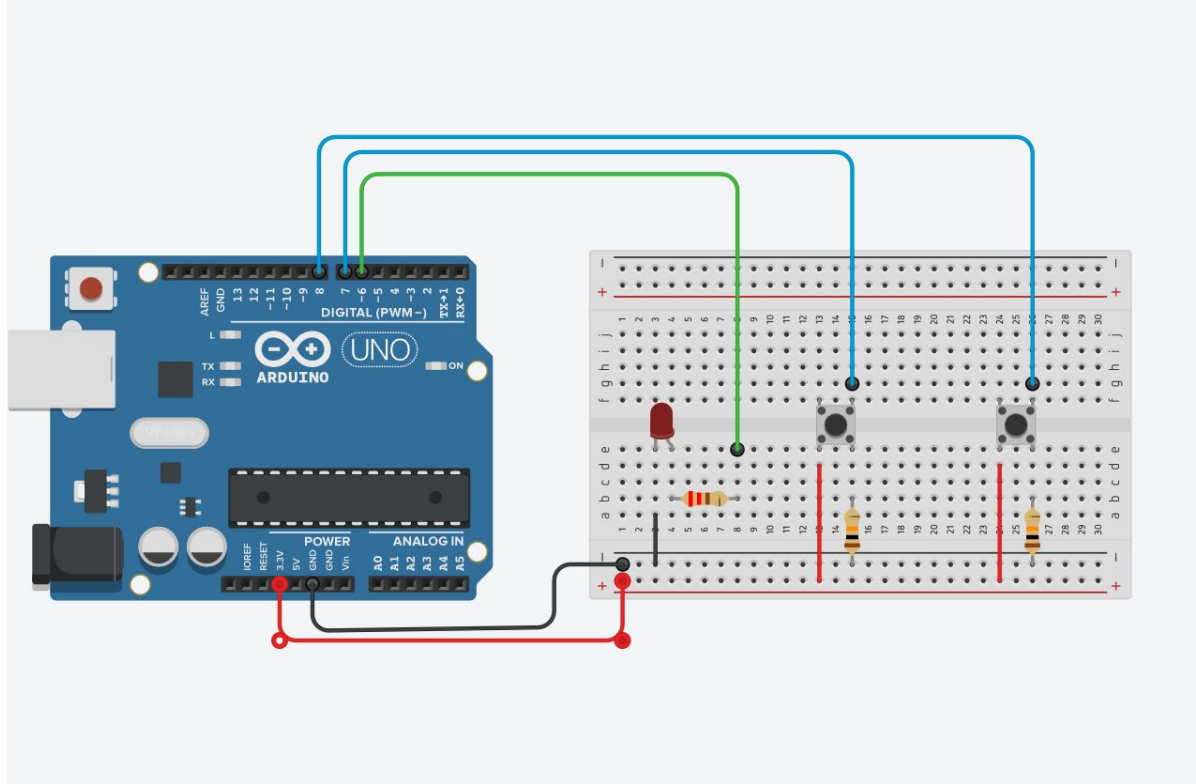
Fizzbuzz! Seven segment

Reaction timer

tinyurl.com/luarduino

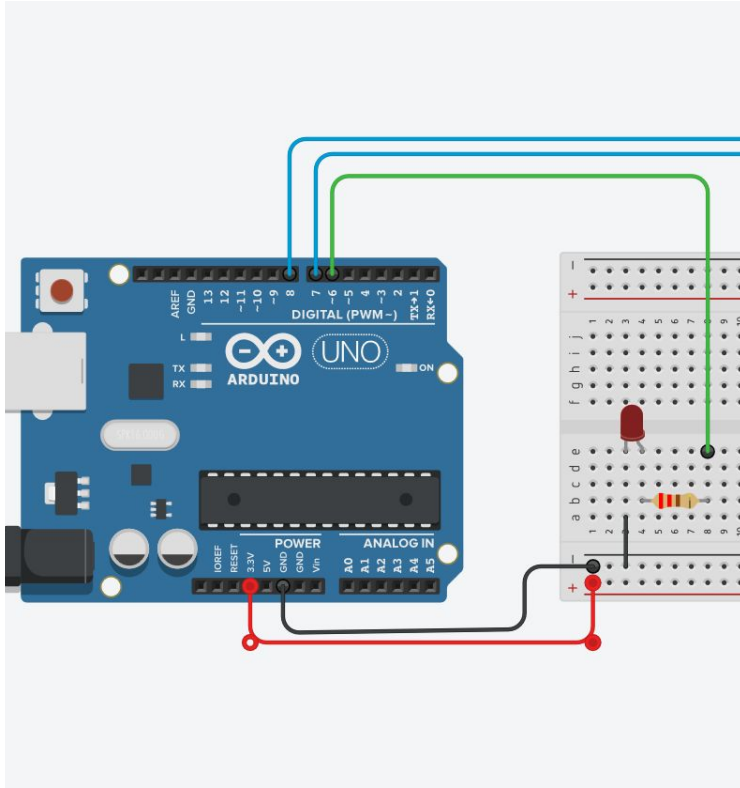
Logic Gates! - making an AND gate!

<https://github.com/LUCompSoc/arduinoExamples> - premade code examples
(try on your own if you're feeling adventurous)



Click through
these slides for
detailed
instructions!

Task 1 - Connect the LED to the arduino.

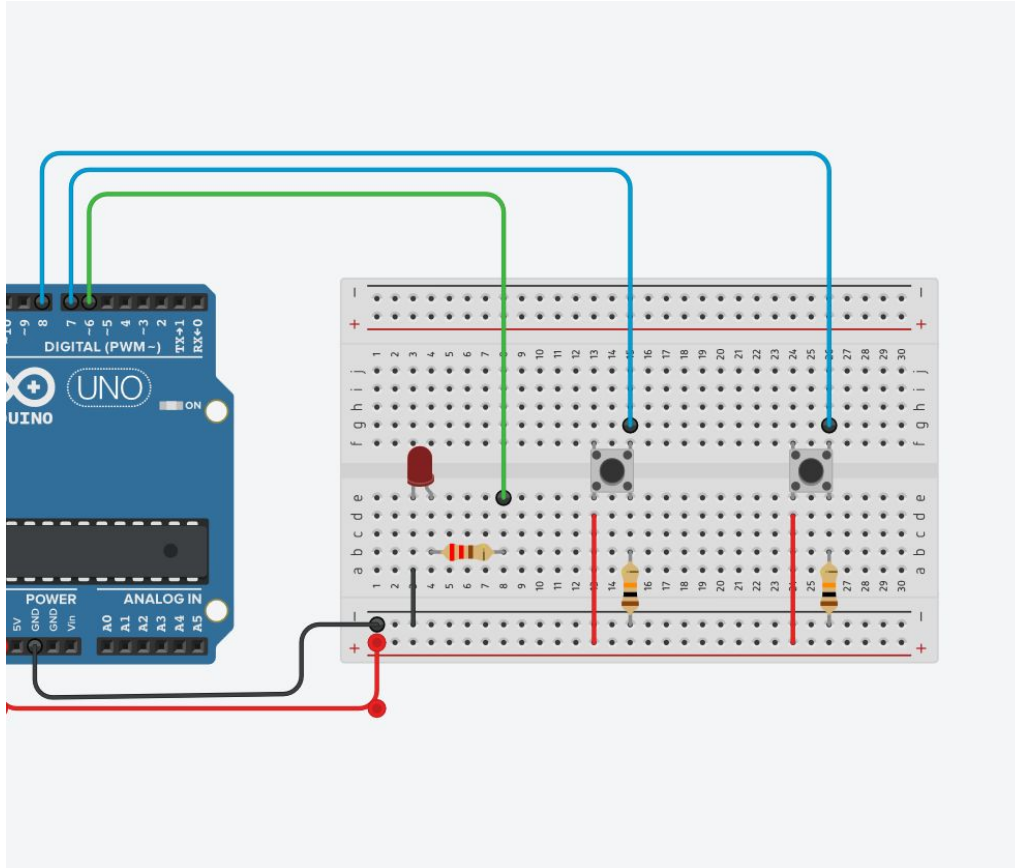


The shorter leg of the LED connects to ground. Use the + - rails on the side of the breadboard to make this easier!

Connect the longer leg to a 220Ω resistor (RED,RED,BROWN,GOLD), then to one of the digital pins (we used pin 6)

The resistors have no polarity - any way around works.

Task 1 - Connect the buttons

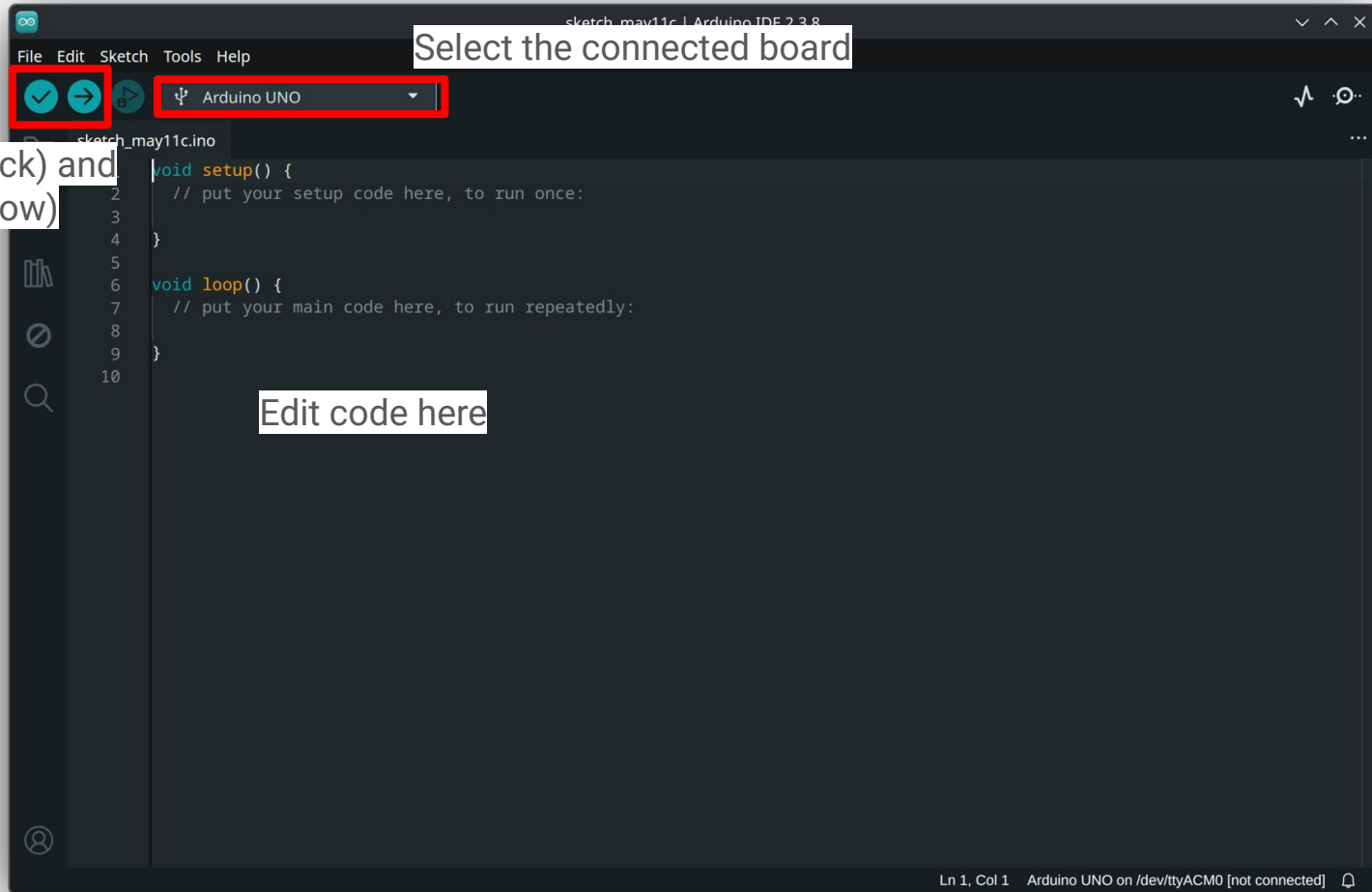


Connect one side of the buttons to 5v using the rail.

Connect the other side to a negative 10KΩ resistor (BROWN,BLACK,ORANGE,GOLD) and the digital pins (we used 7 & 8)

The Arduino IDE

<https://docs.arduino.cc/software/ide/>



Program the arduino!

```
#define BUTTON_A 8  
#define BUTTON_B 7  
#define LED 6  
int buttonAState=0, buttonBState=0;
```

Setup some preprocessor macros and initialise globals to make it easier to work with the code.

Make sure the pin numbers match what you have used in your circuit.

The setup function

```
void setup() {  
    pinMode(LED, OUTPUT);  
    pinMode(BUTTON_A, INPUT);  
    pinMode(BUTTON_B, INPUT);  
}
```

The setup function runs one time, when you power on the arduino.
We need to initialise our pins, so that they are ready to input/output.
Using our preprocessor macro, use pinMode() to set the pins.

The loop function!

```
void loop() {  
    //reads both pins, storing either HIGH or LOW in a variable  
    buttonAState = digitalRead(BUTTON_A);  
    buttonBState = digitalRead(BUTTON_B);  
    //if both buttons are high, our AND gate is true!  
    if (buttonAState == HIGH && buttonBState == HIGH){  
        //So we turn on the LED by writing HIGH to that pin  
        digitalWrite(LED, HIGH);  
    }  
    else{  
        //otherwise we turn it off!  
        digitalWrite(LED, LOW);  
    }  
}
```

Code in our loop function will run repeatedly while the Arduino is powered!

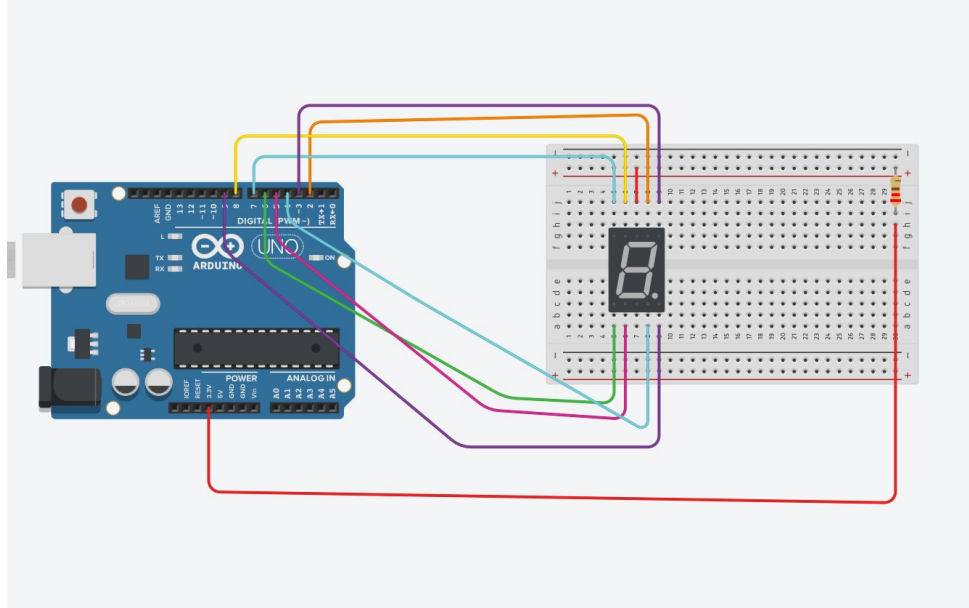
Upload your code!

Select your board, and click the arrow! It should upload and start working if everything is correct.

Once you've done that, try editing your code to make:

- An OR gate
- A NAND gate
- An XOR gate
- A half adder (use logical operators only :p - you'll need an extra led)

Fizzbuzz! - 7 Segment Display!



Wire up the seven segment display!

If you get the pin in the wrong order, don't worry you can change it.

Fizzbuzz! - programming

Fizzbuzz is a counting game. Players go round counting up from 1. If a number is divisible by 3, they say fizz, if it's divisible by 5 they say buzz, if it's divisible by 3 and 5 they say FizzBuzz!

We want to program the seven segment display so it does this! Use the fizzBuzz sample in the git repo, and fill in the loop function to make a working fizzbuzz counter. With special digits in the WriteDigit method it uses: Clear: 10 Fizz: 11, Buzz: 12, FizzBuzz: 13

Try using the template in the repo to get started! Fill in the loop method on your own.

<https://github.com/LUCompSoc/arduinoExamples>

FizzBuzz! Bonus tasks

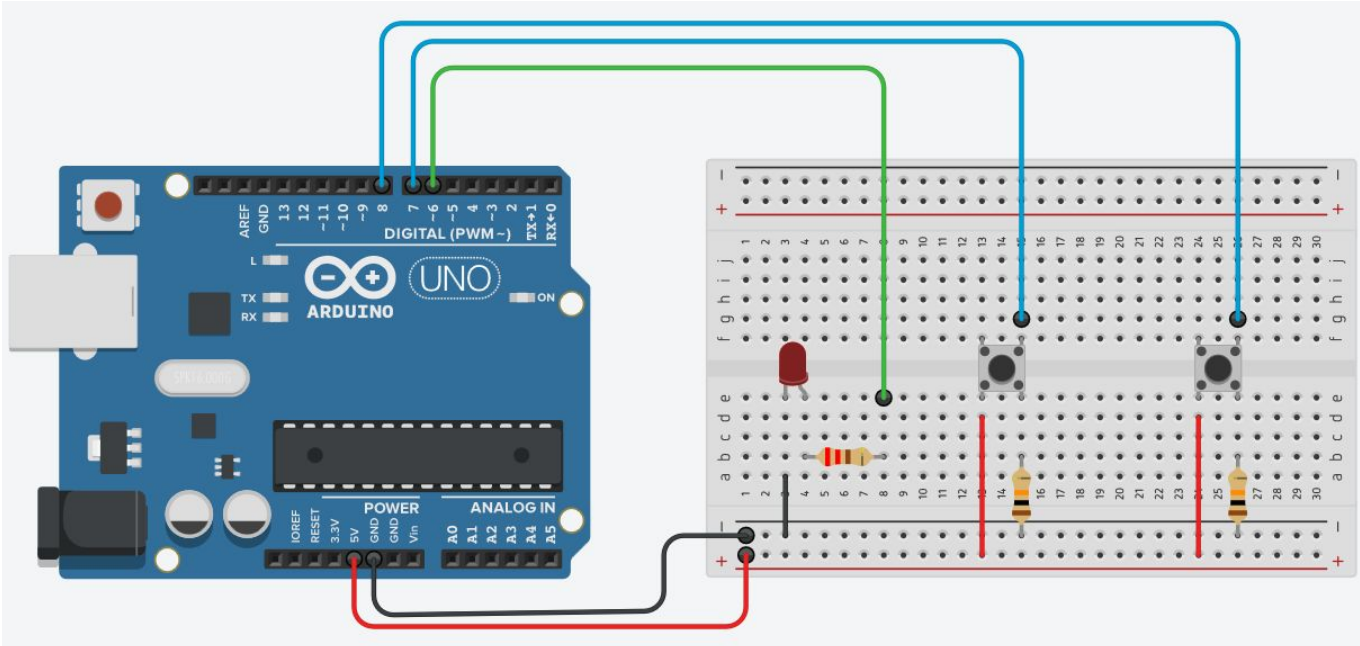
Try adding another 7 segment display (if we have enough :p)

Add a button to increment the counter manually

Add a LED to illuminate when there is a Fizz, Buzz or FizzBuzz.

Try improving the animation shown on screen when there is a fizz or buzz - you could have an animated spinner, or something else!

Reaction timer! - Use the same breadboard setup as for the logic gates



Checkout that section's slides if stuck!

Clock!

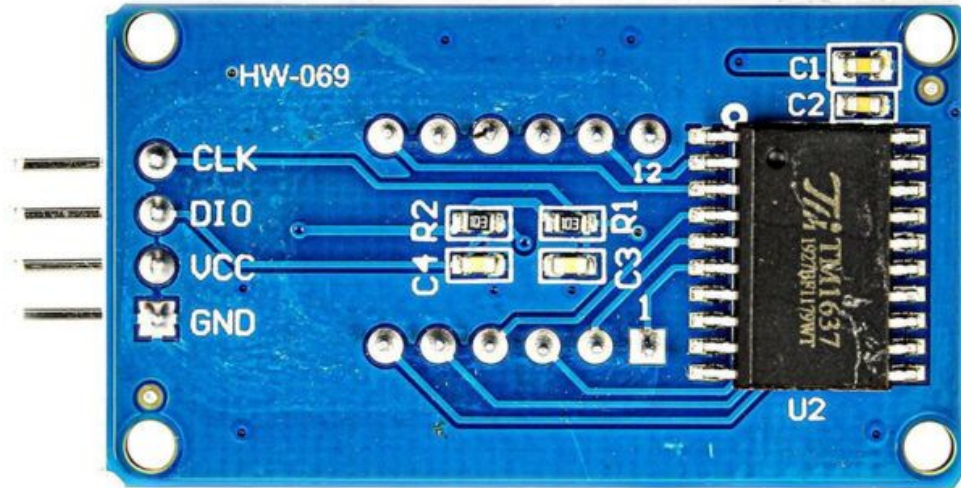
Connect VCC and GND to 5V and GND on your breadboard respectively.

CLK to pin 2

DIO to pin 3

Check out the repo to get started!

<https://github.com/LUCompSoc/arduinoExamples>



Fill in the gaps!

Complete the code in the reactionTimeGame folder in the repo!

You'll need to install the library, following instructions in the README.

Some of the code in the loop function went missing - fill in the missing lines and question marks! If you're stuck, take a look at the other functions.

Try out the timer task as well!

If you want to stretch yourself, add a button and make it a 2 player competitive game!

```
void loop() {
    //If the current time is greater than or equal to the next
    trigger time - replace question mark!
    if (?) {
        //Turn on the LED - fill in the line!
        ?;
        //If the button is pressed - replace the question mark!
        if (?) {
            //Work out how many ms have passed - replace the question
            mark!
            int speed = ?;
            display.showNumberDecEx((speed<9999?speed:0000),
            0b00000000, true);
            delay(1000);
            while (digitalRead(PIN_A)==LOW){
                delay(10);
            }
            //Turn off the LED, clear the display, and generate a new
            trigger time - fill in next 3 lines!
            ?;
            ?;
            ?;
        }
    }
    delay(1);
}
```